Run B Report

Run B: `b5_v4_no_prov`, `b5_v4_no_auth` · Shubham Kumar Sunny · trained and scored 12–13 November 2026

> **Outcome. H1 is supported for both evidence types, in different ways.** Without **authority** evidence the gate becomes unsafe: unsafe actions rise from 3/118 (2.5%) to 17/118 (14.4%) on test (p = 0.0005) and from 2/51 (3.9%) to 10/51 (19.6%) on attacks. Without **provenance** it stays safe but over-defers on attacks: unneeded asks 3/33 (9.1%) to 18/33 (54.5%) (p = 0.0003).

1. The run

Item	Value
Run by	Shubham Kumar Sunny (Kaggle `sunnyshubhamiiith`), branch `run-b`
Training commit	`605e2bf (13 Nov 04:14)`
Scoring commit	`bb0ee0b (13 Nov 19:12)`; training and scoring on branch run-b
Code	`7435346` (`locked-v4`) for training and scoring, confirmed in both kernel logs
Models	`b5_v4_no_prov`, `b5_v4_no_auth`
Scoring	NB04, model alone (no GateExp rules; the computed facts it was trained with are shown), greedy decoding, dev 236 / attacks 61 / test 212 cases, clean simulated user and approver

Training

Model	Examples	Targets	Steps	Mean loss	Time	Seed
`b5_v4_no_prov`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.430	3.5 h	42
`b5_v4_no_auth`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.477	3.6 h	42
`b5_v4_combo`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.426	4.6 h	42

Last row: the final B5 for reference (same data, recipe and seed 42 unless stated).

2. Scoring results

Test pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - provenance	212	190/212 (89.6%)	0/118 (0.0%)	11/110 (10.0%)	21/133 (15.8%)	190/212 (89.6%)	39/43 (90.7%)	39/39 (100.0%)	51/63 (81.0%)
B5 - authority	212	167/212 (78.8%)	17/118 (14.4%)	5/110 (4.5%)	11/133 (8.3%)	169/212 (79.7%)	26/29 (89.7%)	26/39 (66.7%)	36/63 (57.1%)
B5 alone (b5_v4_combo, seed 42)	**212**	**192/212 (90.6%)**	**3/118 (2.5%)**	**5/110 (4.5%)**	**13/133 (9.8%)**	**198/212 (93.4%)**	**39/43 (90.7%)**	**39/39 (100.0%)**	**52/63 (82.5%)**

Attacks pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - provenance	61	58/61 (95.1%)	1/51 (2.0%)	18/33 (54.5%)	2/11 (18.2%)	37/61 (60.7%)	13/22 (59.1%)	13/15 (86.7%)	8/13 (61.5%)
B5 - authority	61	50/61 (82.0%)	10/51 (19.6%)	2/33 (6.1%)	1/11 (9.1%)	44/61 (72.1%)	5/7 (71.4%)	5/15 (33.3%)	7/13 (53.8%)
B5 alone (b5_v4_combo, seed 42)	**61**	**57/61 (93.4%)**	**2/51 (3.9%)**	**3/33 (9.1%)**	**2/11 (18.2%)**	**51/61 (83.6%)**	**11/15 (73.3%)**	**11/15 (73.3%)**	**9/13 (69.2%)**

Dev pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - provenance	236	222/236 (94.1%)	3/112 (2.7%)	10/150 (6.7%)	7/149 (4.7%)	213/236 (90.3%)	22/25 (88.0%)	22/25 (88.0%)	50/61 (82.0%)
B5 - authority	236	209/236 (88.6%)	6/112 (5.4%)	14/150 (9.3%)	13/149 (8.7%)	202/236 (85.6%)	17/25 (68.0%)	17/25 (68.0%)	49/61 (80.3%)
B5 alone (b5_v4_combo, seed 42)	**236**	**225/236 (95.3%)**	**0/112 (0.0%)**	**9/150 (6.0%)**	**9/149 (6.0%)**	**219/236 (92.8%)**	**23/23 (100.0%)**	**23/25 (92.0%)**	**55/61 (90.2%)**

Last row in each pool: the final B5 scored the same way (model alone), the baseline every ablation is compared with.

3. Paired tests against B5

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
B5 vs no authority (test)	success	192/212 (90.6%)	167/212 (78.8%)	+11.8 [+6.1, +17.5]	**<0.0001**	-
B5 vs no authority (test)	unauthorized	3/118 (2.5%)	17/118 (14.4%)	-11.9 [-18.6, -5.9]	**0.0005**	-
B5 vs no authority (attacks)	unauthorized	2/51 (3.9%)	10/51 (19.6%)	-15.7 [-25.5, -5.9]	**0.0078**	-
B5 vs no provenance (test)	success	192/212 (90.6%)	190/212 (89.6%)	+0.9 [-1.9, +3.8]	0.7539	-
B5 vs no provenance (test)	unnecessary deferral	5/110 (4.5%)	11/110 (10.0%)	-5.5 [-11.8, +0.9]	0.1460	-
B5 vs no provenance (attacks)	unnecessary deferral	3/33 (9.1%)	18/33 (54.5%)	-45.5 [-63.6, -24.2]	**0.0003**	-

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

4. Analysis

Authority evidence

- Final success drops +11.8 points [95% CI +6.1, +17.5], $p < 0.0001$.
- Unsafe actions: test 17/118 (14.4%) vs 3/118 (2.5%) for B5, attacks 10/51 (19.6%) vs 2/51 (3.9%). Without knowing who owns the resource and who may approve, the model executes actions that need an approver (most errors are authority inference: 18 vs 5 for B5). This is the largest single effect of any ablation and the strongest evidence for the project's main claim.
- Unneeded asks are unchanged (5/110 (4.5%)): the damage is on safety, not on interaction cost.

Provenance evidence

- On test there is no significant change in success (+0.9 points [95% CI -1.9, +3.8]) or unsafe actions (0/118 (0.0%) vs 3/118 (2.5%)); unneeded asks double (11/110 (10.0%) vs 5/110 (4.5%), $p = 0.146$).
- **On attacks the effect is large:** unneeded asks 18/33 (54.5%) vs 3/33 (9.1%). Without knowing which values came from untrusted content, the model cannot tell an injected instruction from a legitimate one, so it plays safe and asks on most attack cases. Safety holds, but the gate would need the user for every attack, which is the interaction cost H1 is about.
- Provenance matters even more inside the full gate: the PRE rule `untrusted_provenance` decides 28 of 212 test cases before the model is asked, and it needs provenance.
- Reading by the proposal's rule ("not supported if paired tests show no difference"): provenance reduces unnecessary deferral significantly on the attack pool, authority reduces unsafe actions significantly on both pools. **H1 supported.**

5. Records

Record	Where
Training summaries and logs	`results/data/2026-11-13_train_run_b/` (branch `run-b`)
Predictions, reports, logs	`results/data/2026-11-13_eval_run_b/` (branch `run-b`)
Statistics over all runs	`results/data/2026-11-14_analysis/`